
Key Findings & Recommendations

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1. Key Findings

1.1 Finding 1: A verified classical baseline (Phase 1)

Our 9-layer tanh PINN reaches relative $L^2 = 2.12 \times 10^{-4}$ on hard-viscosity Burgers ($\nu = 0.01/\pi$), **three times better** than the Raissi et al. target of 6.7×10^{-4} and below the 10^{-3} acceptance threshold. Reaching this required five documented fixes: a Cole-Hopf analytical reference, Latin-hypercube collocation, scipy L-BFGS-B with Raissi's exact settings, Xavier-uniform initialization, and float64 precision. A weak baseline would have made every later quantum claim suspect, so this step is the foundation of the comparison.

1.2 Finding 2: No easy-regime accuracy advantage (Phase 2, 5-seed paired t -test)

At exact parameter parity, the QAPINN and its classical twin are statistically indistinguishable on easy-viscosity Burgers ($\nu = 0.05/\pi$) at $n = 3$ and $n = 4$ ($p > 0.05$). At $n = 5$ the twin actually wins ($p < 0.05$). On the heat equation they tie. We report honestly that the QAPINN shows **no accuracy advantage** over a parameter-matched classical network in the smooth regime. This is a stronger and more useful statement than an unqualified success claim, because it was obtained with a matched control and a paired statistical test.

1.3 Finding 3: Entanglement is essential; measurement and depth are secondary (ablation grid)

Removing entanglement collapses the QAPINN to $L^2 \approx 1.045$, meaning it predicts almost zero everywhere and does not solve the PDE, against 0.018 with linear entanglement, a factor of **57×**. All-to-all entanglement matches or beats linear, with the gap growing at larger width, consistent with Schuld et al.'s theorem on the wider frequency access of fully entangled circuits. On measurement, expectation values (n outputs) outperform the probability-vector readout (2^n outputs) at small n by concentrating the gradient signal. On depth, a deeper circuit helps marginally on Burgers but hurts on the smooth heat equation, a mild over-parameterization effect. The ranked importance of the design axes is: **entanglement first, then measurement, then depth**.

1.4 Finding 4: The quantum layer accesses the right frequencies, reproducibly

On hard-viscosity Burgers the QAPINN recovers the analytical Fourier centroid to within **1%** for $n \geq 4$. Both the QAPINN and its twin recover about 99% of the true high-frequency content, so the quantum layer reliably represents the correct spectrum, but on this benchmark that reach is not something the classical twin fails to achieve. The difference that does survive is reproducibility: the QAPINN's seed-to-seed spread on the recovery metric is about **2.8×** smaller than the twin's and **19×** smaller than GAAF-PINN's. We read this as a frequency-access result that is **reliable and reproducible** rather than a raw reach the classical network lacks.

1.5 Finding 5: Adding qubits costs gradient signal and exponential time

McClean's theorem predicts an exponential gradient decay for deep random circuits. Our circuits are shallow ($L = 6$) and structured, so the raw gradient variance does not collapse to the strict 3^{-n} floor; the values stay in the range 0.11 to 0.49 across $n = 3$ to 7. However, the matched design reveals a subtler and real effect: across the full width range the QAPINN's first-layer gradient variance decays by about **14×** from $n = 3$ to $n = 8$, while its parameter-matched classical twin stays flat (a factor of 1.03). A decay that the matched classical control does not show is quantum-specific by construction. The per-step wall-clock time of the state-vector simulation roughly doubles per qubit, growing **28×** from $n = 3$ to $n = 8$ and reaching **965×** the twin's cost at $n = 8$. The central practical message: **adding qubits past the target's spectral support buys no accuracy, costs gradient signal, and costs exponential time**.

1.6 Finding 6: The quantum layer is not cosmetic, and it cannot saturate

CKA lets us ask whether the quantum layer actually shapes the prediction or is a passive front end. Its first-layer-to-output alignment is well above zero at every configuration (in the range 0.42 to 0.92 on hard Burgers), so the quantum layer is **not cosmetic**. The clearest and most robust Workstream-C result concerns saturation.

Because each quantum feature is a bounded expectation $\langle\sigma_z\rangle \in [-1, 1]$ and a value beyond 0.95 would require a near-exact computational eigenstate, the quantum first layer registers **exactly zero saturated inputs** at every width and time, whereas the classical tanh front layer it replaces saturates on 5.5% of inputs at $\nu = 0.05$ and on 12.9% at $\nu = 0.01/\pi$. The quantum readout has a **structural protection against saturation** that the classical front layer lacks, and the gap widens on the harder benchmark. This is a concrete, mechanistic answer to the challenge’s “how” question.

2. Recommendations to BQP

2.1 Recommendation 1: Use a narrow, fully entangled, re-uploading circuit with expectation-value readout

The validated core recipe is: angle encoding with data re-uploading, all-to-all entanglement, local $\langle\sigma_z\rangle$ measurement, and a deep classical head (5 layers of width 20). The qubit count should be **matched to the target’s spectral support**, not set to a default maximum. On hard-viscosity Burgers the minimum L^2 is at $n = 6$, not $n = 8$; beyond that point, accuracy does not improve while cost grows exponentially.

2.2 Recommendation 2: Apply a spectral gate before committing to a quantum layer

If the target PDE solution is dominated by low frequencies (for example, the heat equation), a classical PINN is preferable and cheaper. If the solution has genuine high-frequency content (for example, a Burgers shock or a sharp interface), the QAPINN’s frequency-access and non-saturation advantages become relevant. This is the theoretical basis of the qubit-sizing rule: width should follow the target’s frequency content, not a maximum-budget heuristic.

2.3 Recommendation 3: Always compare against a parameter-matched classical twin

A raw quantum-versus-classical comparison confounds the quantum mechanism with the parameter budget. Building a twin with identical parameter count is the only way to attribute a measured difference to the quantum layer alone. Run at least five seeds and a paired t -test on $\log L^2$; a single seed can produce a cherry-picked result. This is the backbone of a fair comparison and the only design that turns “the QAPINN is different” into “the quantum layer is the cause”.

2.4 Recommendation 4: Prioritize entanglement in circuit design

Entanglement is the single largest design lever, with a $57\times$ error swing between the best and worst ablation configurations. All-to-all entanglement is at least as good as linear nearest-neighbor, with the gap growing at larger width. A product-state ansatz (no entanglement) is never competitive and should not be considered for production use.

2.5 Recommendation 5: Do not add qubits past the target’s frequency support

Accuracy does not improve monotonically with width. On hard-viscosity Burgers the QAPINN’s L^2 drops from 0.118 at $n = 3$ to 0.017 at $n = 6$, then rises to 0.053 at $n = 7$. Beyond the spectral support of the target, you pay exponential wall-clock cost for no accuracy gain and decaying gradient signal. The cost per step grows roughly 2^n , reaching $965\times$ the twin’s cost at $n = 8$.

2.6 Recommendation 6: Next steps for production CFD readiness

(a) Run the full 5-seed paired comparison at hard viscosity, where the QAPINN shows the most promise. (b) Test on a second nonlinear PDE such as Allen-Cahn, which has a sharper interface than Burgers, to confirm the frequency-selective interpretation. (c) Move to GPU-accelerated simulation via `lightning.gpu` to scale beyond 8 qubits, where barren plateaus should sharpen. (d) Ultimately, validate on a real 2-D Navier-Stokes benchmark, since every result here is 1+1D and the frequency-selective interpretation is not yet demonstrated on production CFD.